

Counterparty Risk Optimization

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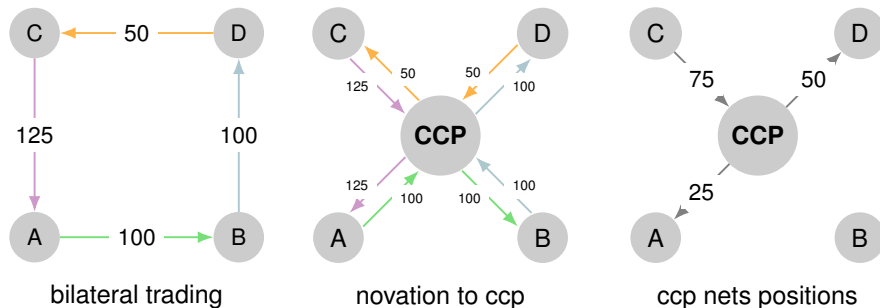
- Risk Measures
 - Types of Risk
 - Central Clearing
 - Value at Risk (VaR) and Expected Shortfall (ES)
 - Initial Margin
- Optimization
 - Optimizing Initial Margin within a Network
 - Challenges we run into in Real Life

- **market risk:** arises from market movements
- **credit risk:** arises from debtors non-payment of an obligation
- **operational and legal risk** arises from people, systems and events
- **liquidity risk** arises when transaction cannot be executed at market price or from an inability to fund outgoing cashflows
- **counterparty risk** arises from the potential exposure of the market value of transactions when a counterparty defaults

Mitigation of Counterparty Risk

- **netting** : net together offsetting payments
- **hedging** : create new trades to offset the risk
- **collateralisation**: collection of collateral or margin payments
- **other contractual clauses** such as resetting mark-to-market (mtm) value or break clauses
- **central clearing counterparties (CCPs)**

Central Clearing Counterparties (CCPs)



The CCP facilitates offsetting positions to be netted against each other

intuition: total counterparty risk $\sim \sum_{counterparties} ||position||$

Risk should measure

- *size* of the potential loss
- *probability* of that loss occurring

A function $\rho : X \rightarrow \mathbf{R}$ is *coherent* if it satisfies the following properties

- **normalised** $\rho(0) = 0$
- **homogeneity** $\rho(\alpha X) = \alpha \rho(X)$
- **translation invariance** if A is a deterministic portfolio with guaranteed return α then $\rho(X + A) = \rho(X) + \alpha$
- **monotonicity** if $X_1 \leq X_2$ in all scenarios then $\rho(X_1) \leq \rho(X_2)$
- **sub-additivity** $\rho(X_1 + X_2) \leq \rho(X_1) + \rho(X_2)$

Value at Risk and Expected Shortfall

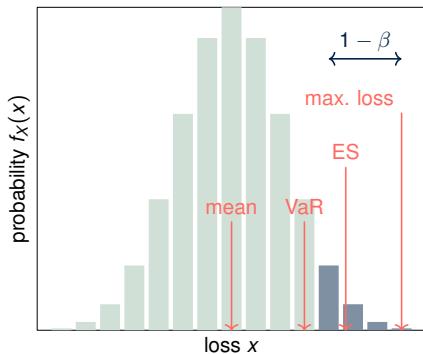
Value at Risk (VaR) is the maximum expected loss at a specific confidence level(β) over some time horizon

$$\int_{-\infty}^{VaR_{\beta}} f_X(x) dx = \beta$$

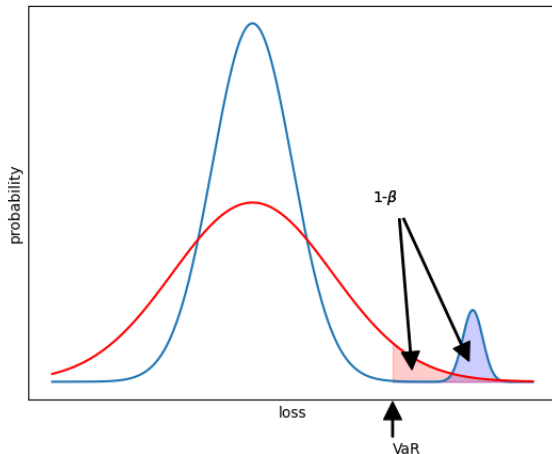
Expected Shortfall (ES) is the expected loss conditional on having exceeded VaR

$$ES = \frac{1}{1 - \beta} \int_{VaR_{\beta}}^{\infty} x f_X(x) dx$$

$$ES \geq VaR$$



fat tails



both portfolios have the same VaR but the blue portfolio has a higher probability of a much bigger loss

Sub-additivity of VaR and ES

A		B	
Loss	Probability	Loss	Probability
100	4%	100	4%
0	96%	0	96%
$VaR_{0.95} = 0$		$VaR_{0.95} = 0$	
$ES_{0.95} = 80$		$ES_{0.95} = 80$	

A+B	
Loss	Probability
200	0.16%
100	7.68%
0	92.16%
$VaR_{0.95} = 100$	
$ES_{0.95} = 103.2$	

- VaR is not sub-additive
- ES is sub-additive

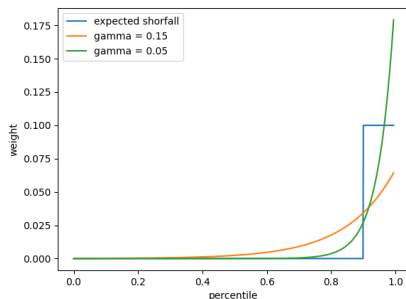
(at least in this case)

easy to find
contrived examples
where VaR is not
sub-additive

hard to *prove* that
Expected Shortfall
is sub-additive

Spectral Risk Measures

A risk measure can be characterised according to the weights it assigns to percentiles of the loss distribution



- VaR assigns 100% to β percentile
- ES assigns equal weight to all percentiles $> \beta$
- exponential spectral risk $W = e^{-(1-q)/\gamma}$

a risk measure is sub-additive if *weight* is a non decreasing function of *percentile*

Time Horizon for VaR and ES

- $\Delta P_i \sim N(\mu, \sigma) \Rightarrow$

$$VaR = \mu + \sigma N^{-1}(\beta)$$

$$ES = \mu + \sigma \frac{e^{-Y^2/2}}{\sqrt{2\pi}(1 - \beta)}$$

where Y is the β percentile point of a standard normal distribution $N(0, 1)$

- T-day $VaR = 1\text{-day } VaR * \sqrt{T}$
- T-day $ES = 1\text{-day } ES * \sqrt{T}$
- If the correlation between ΔP_i and ΔP_{i-1} is ρ , then variance of $\Delta P_{i-1} + \Delta P_i$ is

$$\sigma^2 + \sigma^2 + 2\rho\sigma^2 = 2(1 + \rho)\sigma^2$$

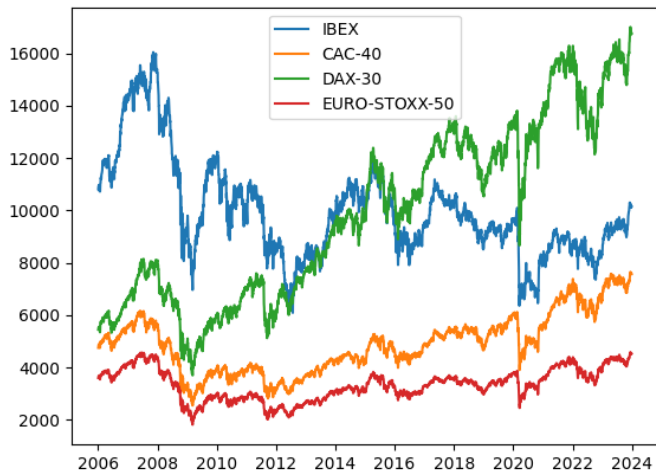
	T=1	T=2	T=5	T=10	T=50	T=250
$\rho = 0$	1.00	1.41	2.24	3.16	7.07	15.81
$\rho = 0.05$	1.00	1.45	2.33	3.31	7.43	16.62
$\rho = 0.1$	1.00	1.48	2.42	3.46	7.80	17.47
$\rho = 0.2$	1.00	1.55	2.62	3.79	8.62	19.35

effect of autocorrelation on ratio of T-Day VaR (ES) to One-Day VaR (ES)

	t	t+1	t+2	t+3
t	1.00	0.12	-0.01	-0.02
t+1	0.12	1.00	0.12	-0.01
t+2	-0.01	0.12	1.00	0.12
t+3	-0.02	-0.01	0.12	1.00

observed auto-correlations for EURO-STOXX

Historical Simulation



Historical Simulation (II)

v_i^j = value of j^{th} market
variable on day i

δ_j = portfolio weight for j^{th}
market variable

$\Delta P_i = \sum_j \delta_j \frac{v_i^j - v_{i-1}^j}{v_{i-1}^j}$ = change
in value of the portfolio on
day i

	CAC	DAX	IBEX	EUROSTOXX
2006-01-02	4757.54	5451.57	10814.60	3605.95
2006-01-03	4803.23	5496.46	10839.30	3638.42
2006-01-04	4838.52	5523.67	10898.90	3652.46
2006-01-05	4849.67	5526.41	10929.60	3661.65
2006-01-06	4867.15	5537.58	10929.60	3666.99
...	...	...	...	...

v_i^j

CAC	DAX	IBEX	EUROSTOXX
3000.00	4000.00	2000.00	1000.00

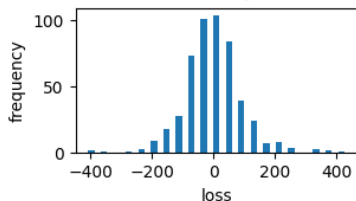
δ_j

Historical Simulation (III)

2022-01-01 -> 2023-12-31

99% VaR=320, ES=377

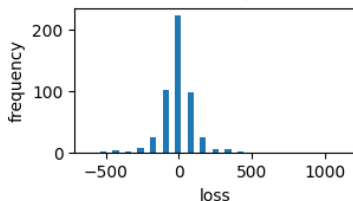
97.5% VaR=227, ES=303



2020-01-01 -> 2021-12-31

99% VaR=450, ES=815

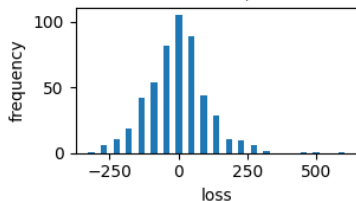
97.5% VaR=306, ES=534



2015-01-01 -> 2016-12-31

99% VaR=297, ES=477

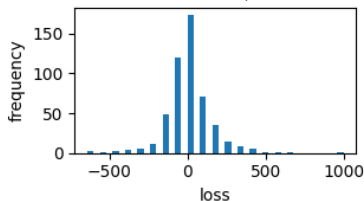
97.5% VaR=246, ES=348



2007-01-01 -> 2008-12-31

99% VaR=448, ES=689

97.5% VaR=370, ES=505



Historical Simulation (IV)

	2022-01-01	2020-01-01	2015-01-01	2007-01-01
12	226.69	305.96	246.47	369.99
11	229.28	309.30	254.17	372.76
10	235.26	344.38	257.59	380.05
9	237.44	351.35	260.55	382.57
8	239.70	369.55	268.87	392.87
7	239.89	393.51	286.93	400.31
6	316.83	397.21	291.90	421.37
5	320.18	450.41	296.98	448.43
4	341.13	596.02	335.56	489.14
3	362.01	712.32	452.86	592.39
2	370.56	800.91	508.75	654.86
1	436.24	1151.84	612.80	1017.71

biggest 12 losses from 500 in each time period

$$\Delta P = \sum_{i=1}^n \delta_i \Delta x_i \quad \delta_i = \Delta P / \Delta x_i$$

$$\Delta x_i \sim N(0, \sigma_i) \Rightarrow \Delta P \sim N(0, \sigma_P)$$

$$\sigma_P = \sqrt{\sum_i \sum_j \rho_{ij} \delta_i \delta_j \sigma_i \sigma_j}$$

$$\sigma_P^2 = \delta^T \mathcal{C} \delta$$

$$VaR(T, \beta) = \sqrt{T} N^{-1}(\beta) \sigma_P$$

$$ES(T, \beta) = \sqrt{T} \frac{e^{-N^{-1}(\beta)^2}}{\sqrt{2\pi}(1-\beta)} \sigma_P$$

	CAC	DAX	IBEX	EUROSTOXX
CAC	0.000102	0.000097	0.000080	0.000094
DAX	0.000097	0.000108	0.000082	0.000096
IBEX	0.000080	0.000082	0.000089	0.000076
EUROSTOXX	0.000094	0.000096	0.000076	0.000096

variance covariance $\mathcal{C}$ matrix for indices based on period
2022-01-01 to 2023-12-31

$$VaR(1, 99\%) = 225 \quad ES(1, 99\%) = 257$$

Probability of exceeding VaR on m or more days from a sample of n is

$$\sum_{k=m}^n \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

$$n = 500, m = 12, p = 0.01 \Rightarrow P = 5.2\%$$

Fundamental Review of the Trading Book

- **Basel I** based on VaR(10 days, 99% confidence)
- **Basel II.5** based on Stress VaR
- **FRTB** based on ES(variable horizon, 97.5% confidence)
 - $99\% VaR = \mu + 2.326\sigma$
 - $97.5\% ES = \mu + 2.338\sigma$

Expected Shortfall as an Optimization Problem

- $X \in \mathbb{R}^n$ is the set of available portfolios
- $Y \in \mathbb{R}^m$ is the set of possible market changes
- $f(x, y) \in \mathbb{R}$ is the loss associated with a given portfolio $x \in X$ and given market change $y \in Y$
- $\rho(y)$ is the density of the probability distribution of y
- $\psi(x, \alpha) = \int_{f(x, y) \leq \alpha} \rho(y) dy$ is the probability of $f(x, y)$ not exceeding α

with above definitions, we can rewrite

- $VaR_\beta(x) = \min\{\alpha \in \mathbb{R} \text{ s.t. } \psi(x, \alpha) \geq \beta\}$
- $ES_\beta(x) = \frac{1}{1-\beta} \int_{f(x, y) \geq VaR_\beta(x)} f(x, y) \rho(y) dy$

Expected Shortfall as an Optimization Problem II

$$F_{\beta}(x, \alpha) = \alpha + \frac{1}{1 - \beta} \int_{y \in \mathbb{R}^m} [f(x, y) - \alpha]^+ \rho(y) dy$$

where

$$[t]^+ = \max(t, 0)$$

- As a function of α , F is convex and continuously differentiable
- $ES_{\beta} = \min_{\alpha \in \mathbb{R}} F_{\beta}(x, \alpha)$

and the portfolio $x \in X$ with the minimum ES can be found by solving the optimization problem

$$\min_{(x, \alpha) \in X \times \mathbb{R}} F_{\beta}(x, \alpha)$$

Expected Shortfall as an Optimization Problem III

$$F_{\beta}(x, \alpha) = \alpha + \frac{1}{1 - \beta} \int_{y \in \mathbb{R}^m} [f(x, y) - \alpha]^+ \rho(y) dy$$

$$\frac{\partial}{\partial \alpha} F_{\beta}(x, \alpha) = 1 + \frac{1}{1 - \beta} (\psi(x, \alpha) - 1) = \frac{1}{1 - \beta} (\psi(x, \alpha) - \beta)$$

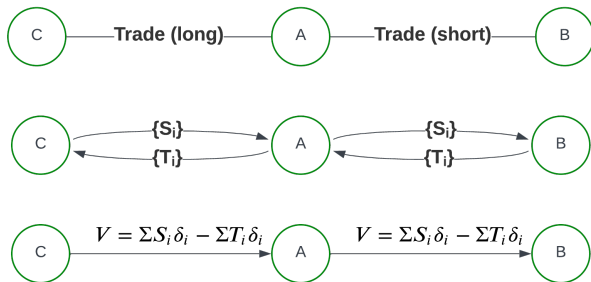
evaluating F_{β} at the point where $\psi(x, \alpha) = \beta$. The integral equals

$$\begin{aligned} \int_{f \geq \alpha_{\beta}} [f(x, y) - \alpha_{\beta}(x)] \rho(y) dy &= \int_{f \geq \alpha_{\beta}} f(x, y) \rho(y) dy - \alpha_{\beta}(x) \int_{f \geq \alpha_{\beta}} \rho(y) dy \\ &= (1 - \beta) ES_{\beta} - \alpha_{\beta}(1 - \psi(x, \alpha_{\beta})) \end{aligned}$$

So

$$\min_{\alpha \in \mathbb{R}} F_{\beta}(x, \alpha) = \alpha_{\beta} + \frac{1}{1 - \beta} [(1 - \beta) ES_{\beta} - \alpha_{\beta}(1 - \beta)] = ES_{\beta}(x)$$

Counterparty Risk and Margin for Derivatives



To mitigate the risk that a counterparty may not pay S_{next} or T_{next} at the next cashflow payment date, **variation margin** $VM = V$ is collected on a daily basis. If a counterparty defaults then they stop posting VM . During the time taken to re hedge or unwind that position, the value V moves due to market movements. This potential loss is covered by collecting **initial margin** (IM)

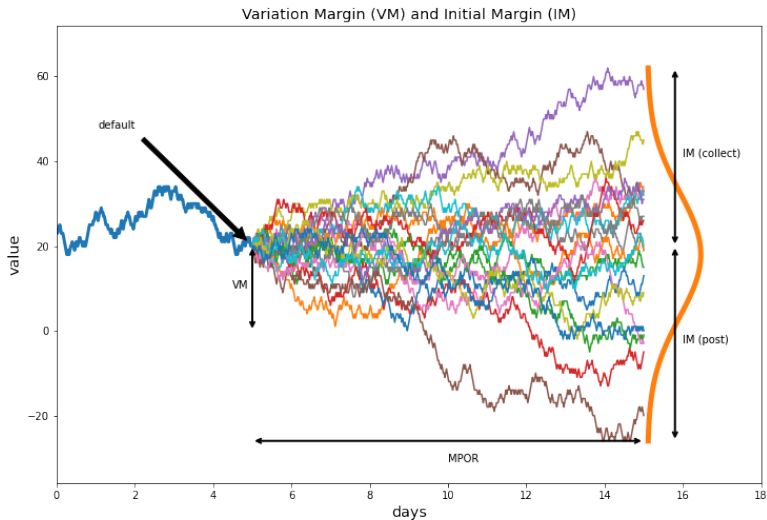
Margin is a form of collateral posted between counterparts to cover potential losses from derivatives

- *Variation Margin* (VM) covers current present value
- *Initial Margin* (IM) covers potential valuation changes due to market movements between when a counterpart default and when the party can exit the position, at time $t = \delta t$

δt is called the *Margin Period of Risk* (MPOR)

Margin is *posted* by the counterparty to the party which *collects* the margin

Margin



Standard Initial Margin Model (SIMM)

- covers non centrally cleared (i.e. *bilateral*) derivative contracts
- has been phased in since 2016 as a result of Basel III rules
- is an exposure based proxy for 10-day VaR at 99% confidence i.e.
 $\delta t = \frac{14}{365}, \beta = 0.99$

Assume we have a portfolio facing some counterparty with value $P(t)$ and that counterparty defaults at time $t = 0$.

- variation margin covers the value at $t = 0$ i.e. $VM(0) = P(0)$
- assume change in P only due to market perturbations
- ignore path dependence, cashflows occurring in $(0, \delta t)$, discounting effects

$$P(\Delta t) \approx P(\mathcal{M}(0) + \Delta \mathcal{M}) \approx P(0) + \delta \Delta \mathcal{M} + \frac{1}{2} \gamma (\Delta \mathcal{M})^2$$

where $\mathcal{M}(t)$ is the state of the market at t and $\Delta \mathcal{M}$ is some perturbation in the market between $t = 0$ and $t = \Delta(t)$

Assume market risk factor movements are jointly Gaussian with zero mean

$$\Delta \mathcal{M} \sim \mathcal{N}(0, \mathcal{C})$$

where $\mathcal{C}_{ij} = \sigma_i \rho_{ij} \sigma_j$ is the covariance matrix.

$P(\Delta t)$ is Gaussian with mean, μ and variance σ^2

$$\mu = P(0) + \frac{1}{2} \text{Tr}(\gamma \mathcal{C})$$

$$\sigma^2 = \delta^T \mathcal{C} \delta + \frac{1}{2} \text{Tr}((\gamma \mathcal{C})^2)$$

trying to find an amount IM such that

$$\mathbb{P}(P(\Delta t) - P(0) < IM) = 0.99$$

which implies

$$IM \approx 2.326 \sqrt{10} \sqrt{\delta^T \mathcal{C} \delta} + \frac{1}{2} \sqrt{10} \text{Tr}(\gamma \mathcal{C}) + \lambda \sqrt{\frac{1}{2} \text{Tr}((\gamma \mathcal{C})^2)}$$

Problems with the formulation so far

- too many risk factors in $\mathcal{M}$. ie $\mathcal{C}, \delta, \gamma$ are too big
- γ is hard to compute

more simplifying assumptions

- $\gamma \propto \frac{\mathcal{V}}{T\sigma_N}$ where $\mathcal{V}$ is the vega of the portfolio
- risk factors (components of $\mathcal{M}$) can be grouped into *buckets* and *risk classes* (eg IR, FX, Equity)
- the portfolio can be separated into product classes with no netting benefit between them
- correlations and other parameters are recalibrated annually
- additional terms address *thresholds* and *concentration risk*.
- some variations of scope between regulators. The posted IM is the maximum SIMM for any regulator

SIMM Tree

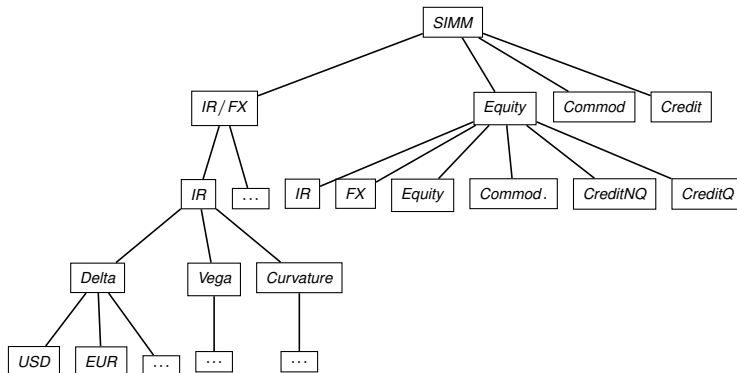
2 : Model

3: Product Class $p \in \mathcal{P}$

4: Risk Class $r \in \mathcal{R}$

5: Sensitivity $s \in \mathcal{S}$

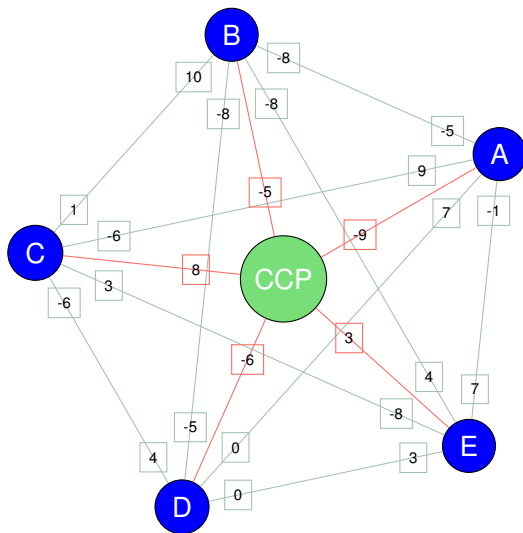
6: Bucket $b \in \mathcal{B}^{s,a}$



Various approaches are used to address (concentrated) positions

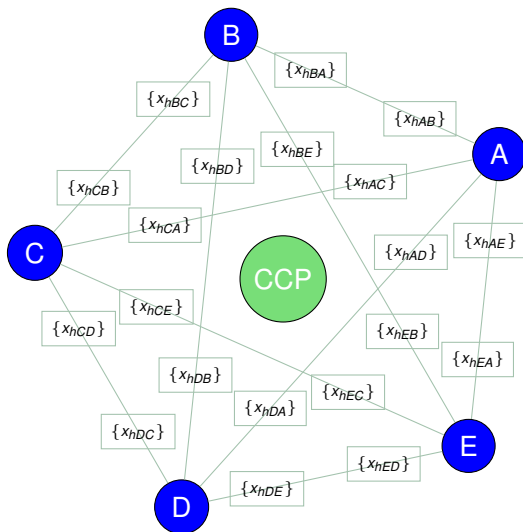
- adjust the Margin Period of Risk $\Delta T = \Delta T_0 \max(1, \frac{N}{N_0})$. For $N > N_0$, $IM \sim N^{\frac{3}{2}}$
- Addons $W(\delta) \delta C \delta^T$ where W is some monotonically increasing function, eg piecewise linear

Multilateral Margin Optimization



network of banks with **cleared** and **bilateral** positions

Multilateral Margin Optimization (II)



goal of multilateral optimization : find hedge trades to reduce the total initial margin in the system

The Margin Minimization Problem

$\mathcal{H}$ is a set of hedges, $\mathcal{M}$ is a set of market risk factors, $\mathcal{P}$ is a set of parties, $\mathcal{D}$ is some risk metric

$$\text{minimize } \sum_{p \in \mathcal{P}} \left(\sum_{q \in \mathcal{P}, q \neq p} SIMM_{p,q} + \sum_{ccp \in CCP} IM_{p,ccp} \right)$$

$$\text{where } SIMM_{p,q} = SIMM_{p,q}(\mathcal{D}_{p,q}), \quad IM_{p,ccp} = IM_{p,ccp}(\mathcal{D}_{p,q})$$

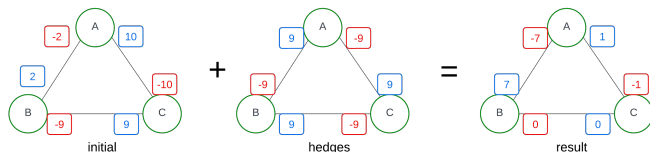
$$\text{such that } \mathcal{D}_{p,q,m} = \mathcal{D}_{p,q,m}^0 + \sum_{h \in \mathcal{H}} x_{h,p,q} \mathcal{D}_{h,p,q,m} \quad \forall m \in \mathcal{M}$$

$$x_{h,p,q} = -x_{h,q,p} \quad \forall h \in \mathcal{H}, \forall p, q \neq p \in \mathcal{P}$$

$$\sum_{q \in \mathcal{P}} x_{h,p,q} = 0 \quad \forall h \in \mathcal{H}, p \in \mathcal{P}$$

$$\mathcal{T}_{p,q,m}^- \leq \sum_h x_{h,p,q} \mathcal{D}_{h,p,q,m} \leq \mathcal{T}_{p,q,m}^+ \quad \forall m \in \mathcal{M}, \forall p, q \neq p \in \mathcal{P}$$

Toy Model of Margin Minimization



Moving a fixed amount of some derivative around a 3-party system is guaranteed to satisfy

- party - counterparty symmetry: $x_{h,p,q} = -x_{h,q,p}$
- cashflow flatness: $\sum_q x_{h,p,q} = 0$

margin $\sum_{p,q} |x_{hpq}^0 - x_{hpq}|$ is minimised when x is the median of x^0

Numerical Solutions of Constrained Optimization Problems

minimize $f_0(x)$ subject to $f_i(x) \leq b_i, \quad i = 1 \dots m$

- solving convex optimization problems is much easier than general non-linear optimization problems
- convexity is equivalent to sub-additivity $f_i(\alpha x + \beta y) \leq \alpha f_i(x) + \beta f_i(y)$
- Minimizing ES is a piecewise linear problem
- *SIMM* is *nearly* convex
- Concentration add-ons are *potentially* convex
- the state of technology for solving convex optimisation problems is very mature
- but the ease of solution is highly dependent on problem specifics

Modelling Languages and Solvers

modelling languages *vastly* simplify the task of setting up an optimization problem

solvers perform the actual numerical steps

several commercial and open source modelling languages and solvers are available

```
import gurobipy as gp
from gurobipy import GRB
m = gp.Model()
x1 = m.addVar(vtype=GRB.INTEGER, name="x1", lb=0)
x2 = m.addVar(vtype=GRB.INTEGER, name="x2", lb=0)
m.setObjective(5*x1 + 8*x2, GRB.MAXIMIZE)
m.addConstr(x1 + x2 <= 5, "c1")
m.addConstr(3*x1 + 7*x2 <= 25, "c2")
m.optimize()
```

Solver output for a simple model

```
Gurobi Optimizer version 9.5.2 build v9.5.2rc0 (linux64)
Thread count: 24 physical cores, 48 logical processors, using up to 24 threads
Optimize a model with 2 rows, 2 columns and 4 nonzeros
Model fingerprint: 0x48b07709
Variable types: 0 continuous, 2 integer (0 binary)
Coefficient statistics:
  Matrix range      [1e+00, 7e+00]
  Objective range   [5e+00, 8e+00]
  Bounds range      [0e+00, 0e+00]
  RHS range         [5e+00, 2e+01]
Found heuristic solution: objective 25.0000000
Presolve time: 0.00s
Presolved: 2 rows, 2 columns, 4 nonzeros
Variable types: 0 continuous, 2 integer (0 binary)
Found heuristic solution: objective 28.0000000

Root relaxation: objective 3.1000000e+01, 2 iterations, 0.00 seconds (0.00 work units)

   Nodes      |   Current Node   |   Objective Bounds      |   Work
 Expl Unexpl |  Obj  Depth IntInf | Incumbent    BestBd   Gap | It/Node Time
*    0       0              0      31.0000000    31.00000    0.00%    -     0s

Explored 1 nodes (2 simplex iterations) in 0.01 seconds (0.00 work units)
Thread count was 24 (of 48 available processors)

Solution count 3: 31 28 25

Optimal solution found (tolerance 1.00e-04)
Best objective 3.1000000000000e+01, best bound 3.1000000000000e+01, gap 0.0000%
```

FX Delta Margin

- 25 parties
- 30 currencies
- cleared and uncleared Non Deliverable Forwards
- 1 maturity (short dated)
- $\sim 18,000$ variables
- $\sim 40,000$ constraints
 - pairwise pointwise risk (linear inequality)
 - pairwise IM increase (convex inequality)
 - notional efficiency (convex inequality)
 - symmetry (linear equality)
 - flatness (linear equality)
 - lot sizes (round notionals to nearest million) (integer)
- 1 convex objective : SIMM + Cleared IM

FX Delta Margin - Numerical Complexity

- inner-most step is Newton step
- inverting $\nabla^2 f$ is the limiting step
- matrix inversion $\sim n^3$ where $n \sim 40,000$ (10^{13} FLOPS)
- in fact we can do *far* better than this
 - positive semi definite $\implies$ Choleski Factorization
 - sparsity ($\ll 0.1\%$ of matrix is non-zero)
 - structure
 - small number of constraints bind multiple trades
 - vast majority (36,000) of constraints are risk constraints
 - each NDF trade only has exposure to 1 risk factor, implying a (block) diagonal (or banded) matrix

Block Matrix Structure

trades

$h_{A,B}$ $h_{A,C}$ $h_{B,A}$ $h_{B,C}$ $h_{C,A}$ $h_{C,B}$

constraints

symmetry, global constraints

flatness, party constraints

risk, (p, cpty) constraints

Dantzig-Wolfe structure

Nearly Parallel Risk

2 identical trades with exposure, $3E$ and E to 2 risk factors. Scaling everything by $3E$ can lead to unstable solutions

- $\begin{bmatrix} 1 & 1 \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{3} \end{bmatrix}$ has infinite number of solutions $y_1 = 1 - x_1$
- $\begin{bmatrix} 1 & 1 \\ 0.3333 & 0.333 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.333 \end{bmatrix}$ has a single solution $(0, 1)$
- $\begin{bmatrix} 1 & 1 \\ 0.3333 & 0.333 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.3333 \end{bmatrix}$ has a single solution $(1, 0)$
- $\begin{bmatrix} 1 & 1 \\ 0.3333 & 0.333 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.3 \end{bmatrix}$ has a single solution $(-110, 111)$
(infeasible if $x_i \geq 0$)

FX Delta Margin

Step I : solve the continuous problem

Threads: 48 physical cores, 96 logical processors, 32 threads

Optimize a model with 114127 rows, 104378 columns

and 618202 nonzeros

Model fingerprint: 0xc3d79de8

Model has 661 quadratic constraints

Coefficient statistics:

Matrix range [4e-04, 1e+01]

QMatrix range [3e-01, 1e+00]

Objective range [1e-02, 1e+00]

Bounds range [1e-05, 2e+04]

RHS range [1e-05, 6e+03]

Presolve removed 62099 rows and 55441 columns

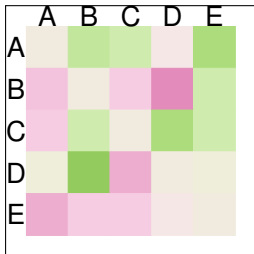
Presolve time: 0.38s

Presolved: 53358 rows, 49577 columns, 473486 nonzeros

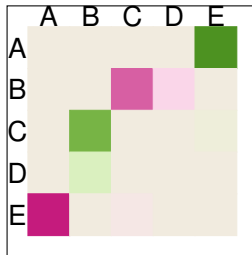
Barrier solved model in 58 iterations and 11.21 seconds

FX Delta Margin (5 parties)

5 party system, 1 risk factor, no constraints



initial risk

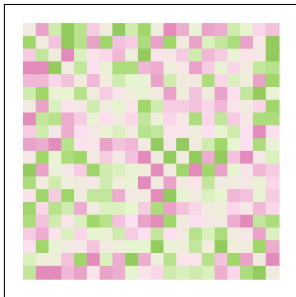


optimised risk

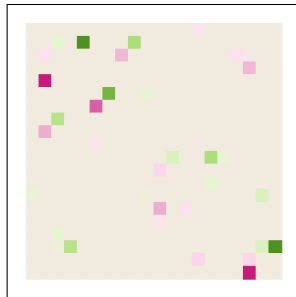
risk on each party is netted. Parties have *either* short or long positions, never both

FX Delta Margin (20 parties)

20 party system, 1 risk factor, no constraints



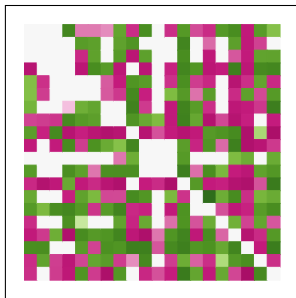
initial risk



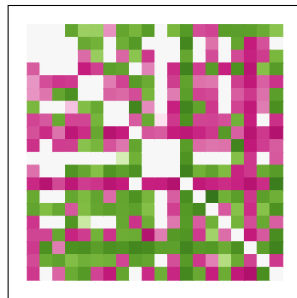
optimised risk

FX Delta Margin (Reality)

A real system with 20 parties and real constraints



initial risk



optimised risk

IR Margin Optimization

- 15 parties
- 3 currencies (USD, EUR, GBP)
- swaptions and cleared swaps
- multiple expiries and maturities
- 5000 variables, 10,000 constraints
- each trade has exposure to multiple risk factors (eg swap expiring in 12 years is exposed to 10Y Swap Rate and 20Y Swap Rate)

Other considerations

- feasibility (*does the solution satisfy all constraints*)
- optimality (*does it find a **good** IM saving*)
- nice
 - round numbers
 - small numbers of trades
 - trade packages
 - centrally cleared vs bilateral
- fair

FX Delta Margin - Rounding

MIP problem so remove as many variables and constraints as possible first

Optimize a model with 49083 rows, 38685 columns
and 116685 nonzeros

Model fingerprint: 0xba7bdcd2

Variable types: 28935 continuous, 9750 integer (0 binary)

Presolved: 6068 rows, 5935 columns, 18230 nonzeros

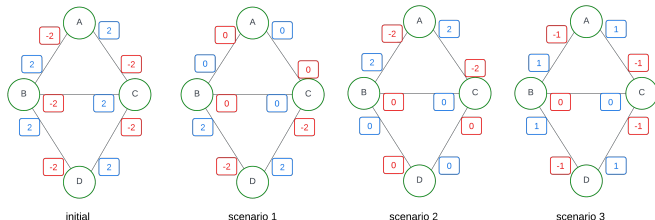
Variable types: 2364 continuous, 3571 integer (30 binary)

Root relaxation: objective 3.348605e+03, 6049 iterations,
0.13 seconds

Explored 5211 nodes (149715 iterations) in 57.36 seconds

Best objective 8.262033036720e+03

Fairness



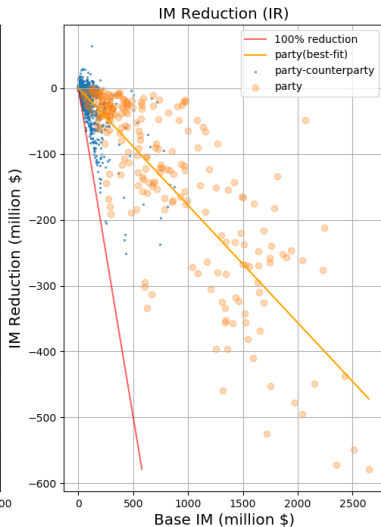
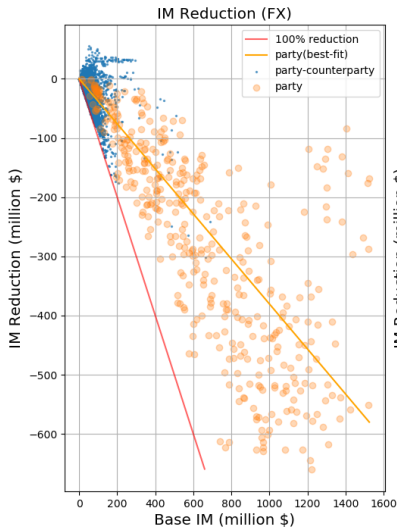
Party	Scenario 1	Scenario 2	Scenario 3	Single Party Best
A	4	0	2	4
B	4	4	4	4
C	4	4	4	4
D	0	4	2	4

3 ways to achieve the same overall margin saving in a 4 party system

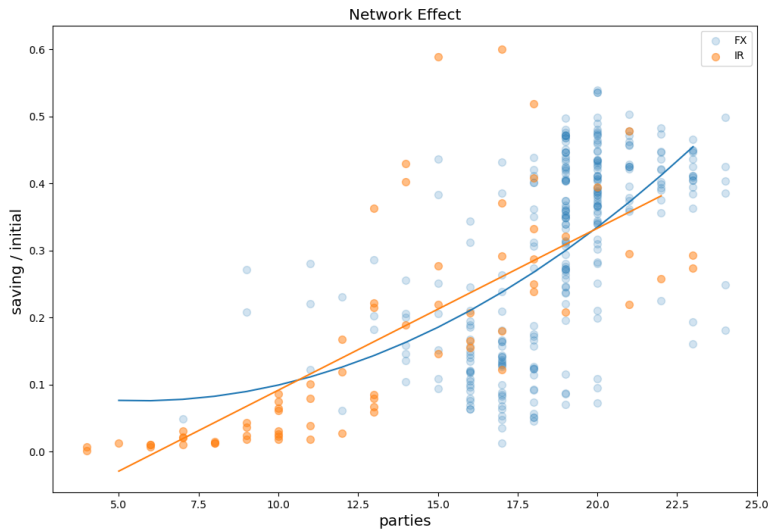
2-step algorithm: compute best possible for each party and then solve

$$\min \sum_p (IM_p - IM_p^{best})^2$$

Observed Margin Savings



Network Effect



Summary

- **Value at Risk** and **Expected Shortfall** are risk measures that describe in a single number the riskiness of a financial portfolio
- **Over the Counter Derivatives** can be **centrally cleared** via a Central Clearing Counterparty or **bilateral**
- trading derivatives creates **counterparty risk** which can be mitigated by the collection of **Variation Margin** and **Initial Margin**
- **Optimization of Initial Margin** within a network of financial participants helps to reduce counterparty risk
- IM Optimization can be achieved via well understood **constrained convex optimization** numerical methods.
- **Numerical Issues** have a significant impact on the difficulty
- Interesting problems arise from making the solution **Nice** and **Fair** as well as Feasible and Optimal

Further Reading

Steven Boyd and Lieven Vandenberghe (2004) **Convex Optimization** Cambridge University Press

Jon Gregory (2020) **The XVA Challenge** (4th edition) Wiley

John Hull (2023) **Risk Management and Financial Institutions** (6th edition) Wiley

Artzner, P., F. Delbaen, J-M. Eber and D. Heath (2022) **Coherent Measures of Risk** *Mathematical Finance* 9(3), 203 – 228

https://www.ise.ufl.edu/uryasev/files/2011/11/CVaR1_JOR.pdf **Optimization of Conditional Value at Risk**

<https://www.bis.org/bcbs/publ/d317.pdf> **Basel III regulatory framework for initial margin for non centrally cleared derivatives**

https://www.isda.org/a/b4ugE/ISDA-SIMM_v2.6_PUBLIC.pdf **ISDA SIMM 2.6**

https://people.math.ethz.ch/~embrecht/ftp/Seven_Proofs.pdf **Seven proofs of the subadditivity of Expected Shortfall**

<https://people.orie.cornell.edu/gennady/techreports/VaRsubadd.pdf>

Subadditivity Re-Examined: the Case for Value-at-Risk

<https://www.isda.org/a/KV9EE/>

[Margin-Requirements-for-NonclearedDerivatives-April-2018-update.pdf](https://www.isda.org/a/KV9EE/Margin-Requirements-for-NonclearedDerivatives-April-2018-update.pdf)

Margin requirements for non cleared derivatives

<https://www.risk.net/our-take/6963251/>

[backtesting-expected-shortfall-mission-accomplished](https://www.risk.net/our-take/6963251/backtesting-expected-shortfall-mission-accomplished) **Backtesting expected shortfall: mission accomplished?**